MIL-S-3786/16B
17 January 1972
SUPERSEDING
MIL-S-3786/16A
5 November 1965

MILITARY SPECIFICATION

SWITCH, ROTARY, ELECTRO-MECHANICAL

ACTUATED (SOLENOID TYPE),

STYLE SR16

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

* DO NOT USE FOR NEW DESIGN
FOR REPLACEMENT PURPOSES ONLY

1. SCOPE

1.1 This specification covers the detail requirements for style SR16 electro-mechanical actuated rotary switches.

2. APPLICABLE DOCUMENTS

2.1 The following document, of the issue in effect on date of invitation for bids or request for proposal, forms a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-S-3786 - Switches, Rotary (Circuit Selector, Low-Current Capacity), General Specification for.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Requirements. Requirements shall be in accordance with MIL-S-3786, and as specified herein.

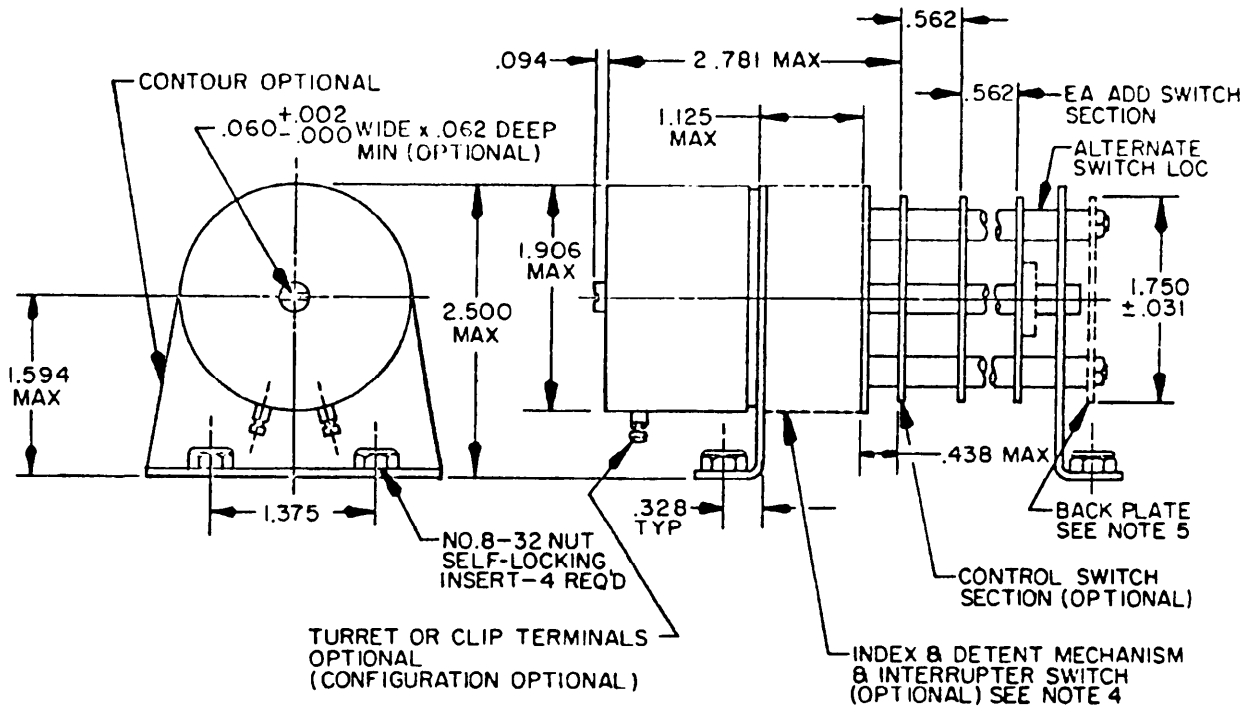
3.2 Design and construction. Switches shall be of the design, construction, and physical dimensions specified on figure 1.

3.2.1 Switch terminals. Terminal tabs shall be bent at an angle of ±15 degrees from the plane of the section except that the common and corner terminals shall be bent at an angle of 45 ±15 degrees from the plane of the section.

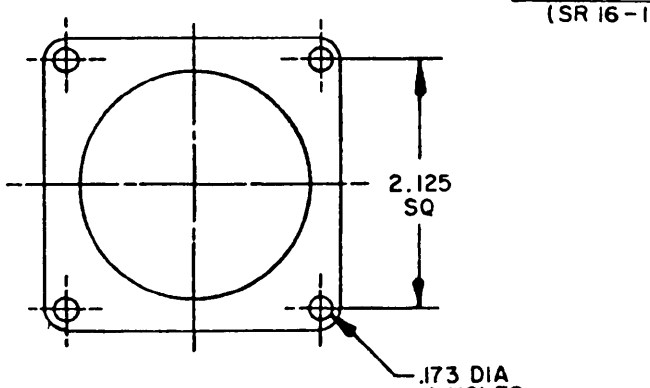
3.3 Solenoid-input voltage. Solenoid-input voltage shall be 24 Vdc minimum, 30 Vdc maximum.

FSC 5930

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FOOT MOUNTING
(SR 16-1)



FLANGE MOUNTING
(SR 16-2)

INCHES	MM
.002	.05
.031	.79
.060	1.52
.062	1.57
.094	2.39
.173	4.39
.328	8.33
.438	11.13
.562	14.27
1.125	28.58
1.375	34.93
1.594	40.49
1.750	44.45
1.906	48.41
2.125	53.98
2.500	63.50
2.781	70.64

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Unless otherwise specified, tolerances shall be ±1/64(.40 mm) on fractions and ±.005 (.13 mm) on decimal dimensions.
4. Interrupter switch mechanism may extend below dotted line, but not below mounting feet.
5. Back plate may be omitted if rear foot bracket is visible and contains markings.

FIGURE 1. Style SR16 solenoid operated switch.

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3.4 Solenoid-input current. Maximum-input shall not exceed 5 amperes at 85°C or 3 amperes at 125°C.

3.5 Contact resistance. Contact resistance shall not exceed the following values:

Initial and after vibration and shock	5 milliohms.
After moisture-resistance and salt spray	5 milliohms.
After life (rotational) tests	10 milliohms.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-3786, and as specified herein.

4.2 Life (rotational). Test loads and applicable-circuit conditions shall be as specified in table I. Each of the loads specified for the applicable-environmental condition shall be switched by at least one rotor contact of the switch.

TABLE I. Circuit values for life (rotational) test.

Environmental condition	Inductive load (2.8 henries)		Resistive loads	
	Milli-amperes	Volts, dc	Milli-amperes	Volts
At atmospheric pressure	50	30	500 50	30 dc 300 ac

4.2.1 Stepping speed. Solenoid-coil current shall be interrupted at a rate sufficient to cause switch deck contacts to be stepped at the rate of 20 steps per second, minimum.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with MIL-S-3786.

6. NOTES

6.1 The notes specified in MIL-S-3786 are applicable to this specification.

6.2 Ordering data. Procurement documents shall specify the following:

(a) Number of switch positions, angle of throw, and electrical configuration.
(b) Homing (control) deck and configuration, as required.
(c) Flange or foot mounting (see figure 1).
(d) Solenoid input voltage range and power source type.
(e) Solenoid duty cycle. Specify maximum "ON" and minimum "OFF" times.
(f) Lead wires and length (when required).

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NOTE: The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Army - MU, MI
Navy - EC, AS, OS
Air Force - 11, 17
DSA - ES

User activities:
Army - ME, AT, AV, SM
Navy - MC, CG
Air Force - 19

Preparing activity:
Air Force - 80

Agent:
DSA - ES

(Project 5930-0830)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

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